Matplotlib Library

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- Chart types
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Simple chart

- Import Matplotlib by Python

```
import matplotlib.pyplot as plt
```

- Declare variables (list)

```
import numpy as np
t = np.arange(0, np.pi, np.pi/12)
s = np.cos(t)                                    # element-wise
```

Simple chart

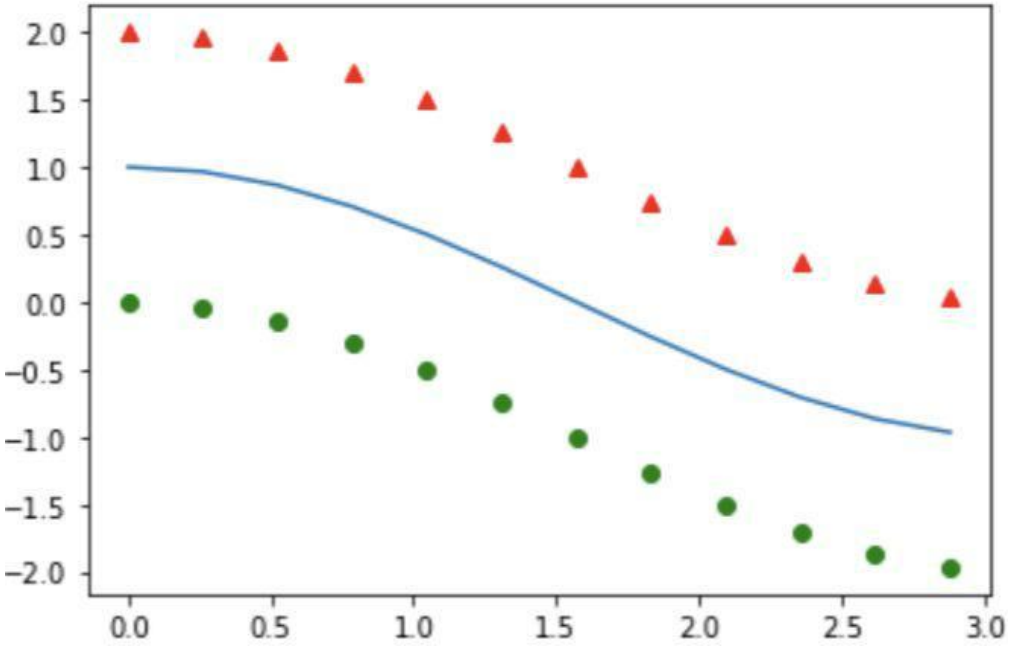
- Plot a line chart or format string (third parameter)

```
plt.plot(t, s)

plt.plot(t, s+1, ‘r^’)

plt.plot(t, s-1, ‘go’)

plt.show()
```



Simple chart

- Format chart

```
plt.plot(t, s)
plt.plot(t, s+1, 'r^')
plt.plot(t, s-1, 'go')
plt.title('example')
plt.xlabel('t (rad)')
plt.ylabel('s')
plt.xlim(0, 3)
plt.ylim(-3, 3)
plt.text(2.5, -0.75, r'$s=cos(t)$')
plt.show()
```

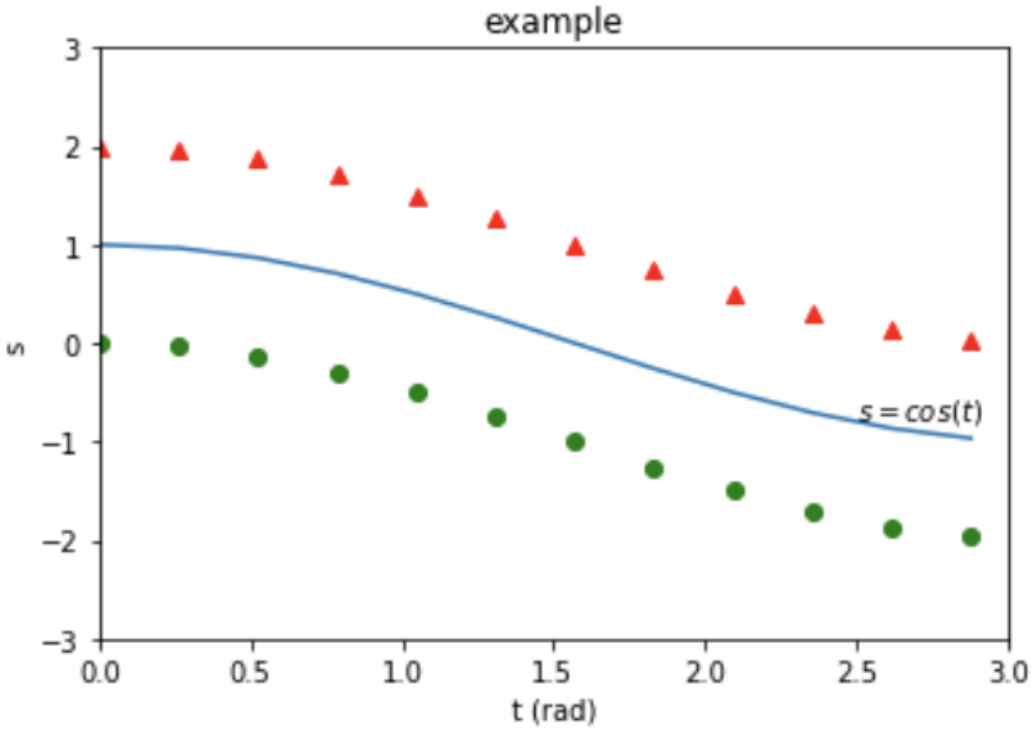


Chart types

● Scatter Plot	plt.scatter()
● Line Chart	plt.plot()
● Bar Chart	plt.bar()
● Pie Chart	plt.pie()
● Stacked Area Chart	plt.stackplot()
● Image	plt.imshow()

Chart types

- Scatter Plot plt.scatter()

```
plt.scatter(t, s+1)
plt.show()
```

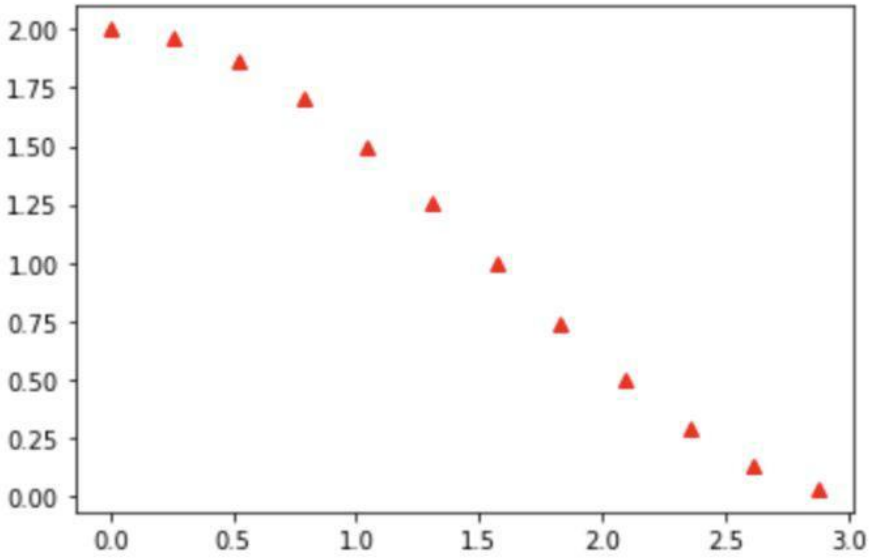


Chart types

- Line Chart

```
plt.plot(t, s+1)
plt.show()
```

plt.plot()

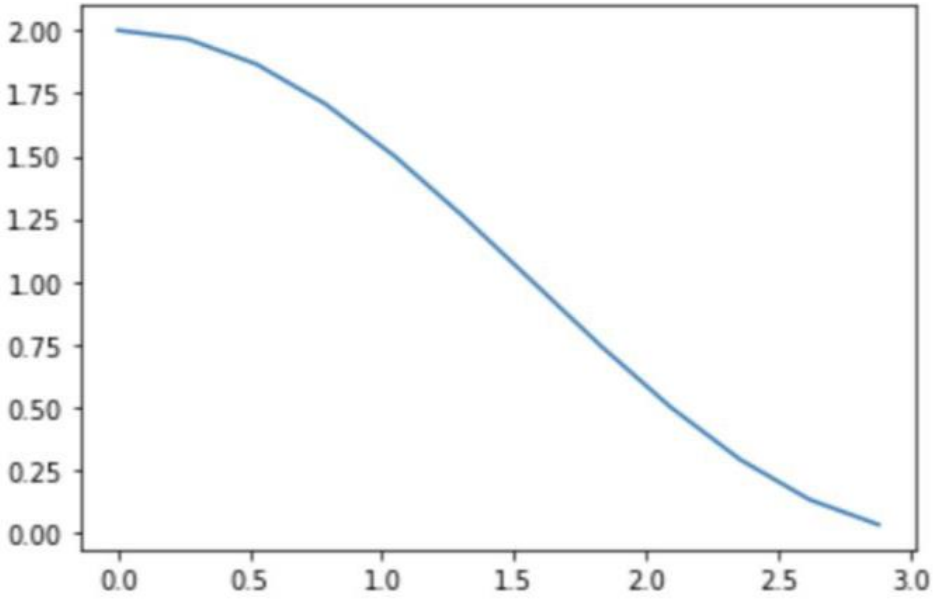


Chart types

- Bar Chart plt.bar()

```
plt.bar(range(3), [30, 45, 50])
plt.xticks(range(3), ['2010',
        '2015', '2020'])
plt.show()
```

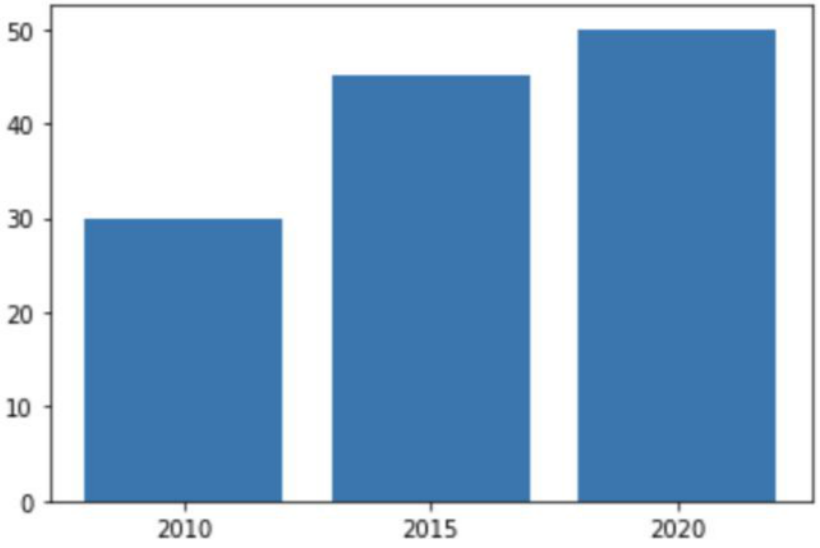


Chart types

- Pie Chart plt.pie()

```
plt.pie([20, 30, 40],
    labels=['A', 'B', 'C'],
    startangle=90,
    autopct='%02d%%')
plt.show()
```

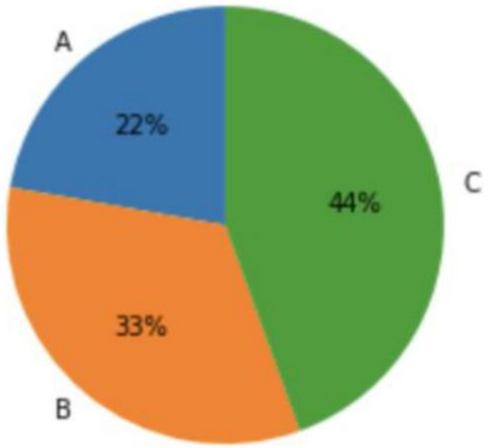


Chart types

- Stacked Area Chart plt.stackplot()

```
x = np.arange(4)
y = x+np.arange(4)
z = y+np.arange(4)
plt.stackplot(x, y, z,
     labels=['A', 'B', 'C'])
plt.show()
```

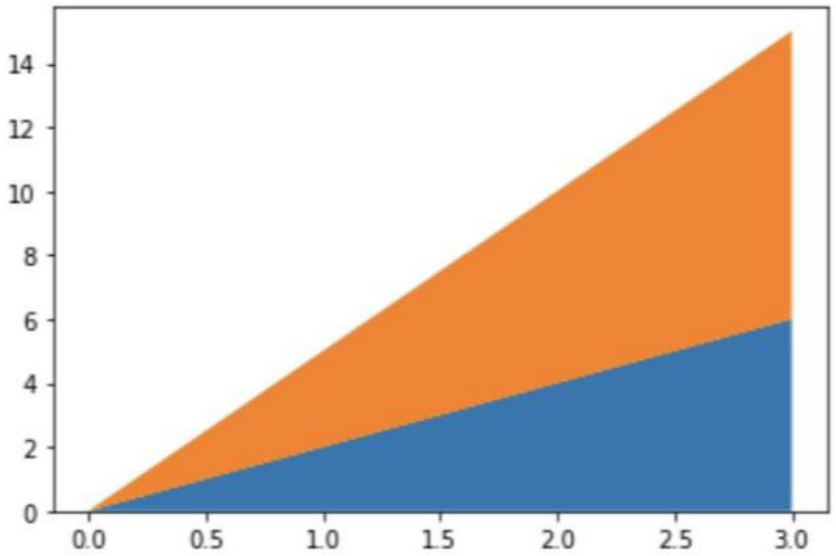
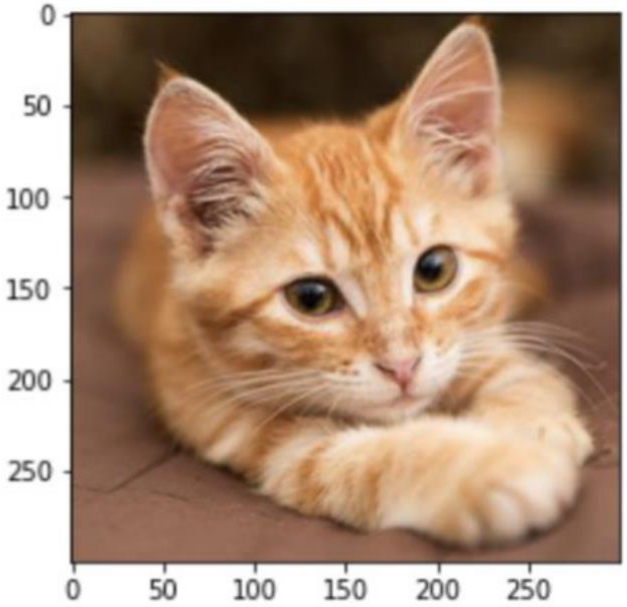


Chart types

- Image plt.imshow()

```
img = plt.imread('a.jpg')
plt.imshow(img)
plt.show()
```



Subplot

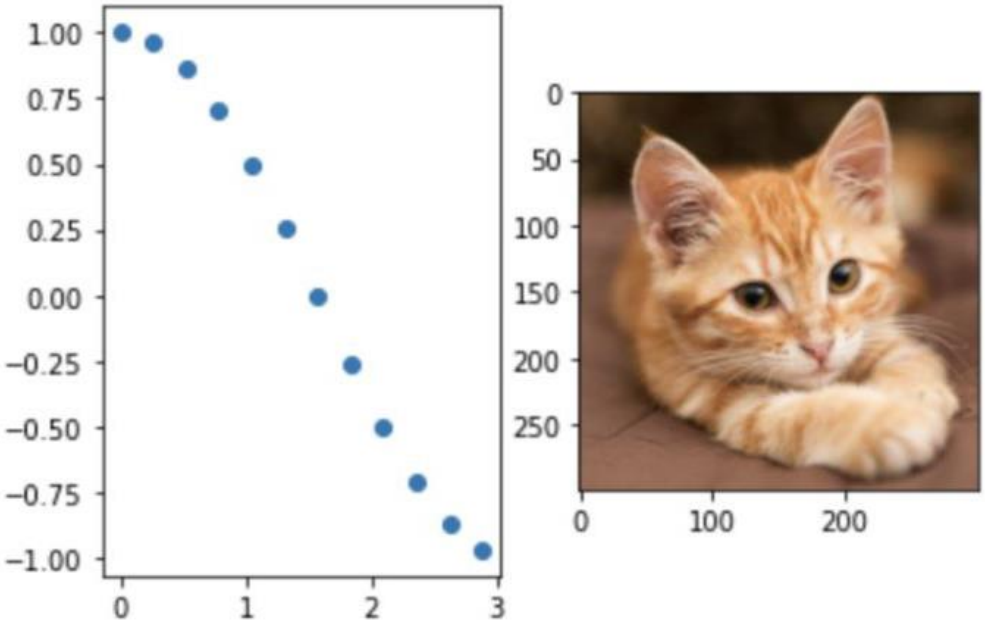
- Position of the subplot:

  plt.subplot(<nrows >, < ncolumns>, <index>)

- Example: plt.subplot(2, 3, 3)

1	2	3
4	5	6

Subplot

- Example

```
plt.subplot(121)
plt.scatter(t, s,
        marker='o')

plt.subplot(122)
img = plt.imread( 'a.jpg')
plt.imshow(img)

plt.show()
```